

East Delta University

School of Science, Engineering & Technology

Title: String

CSE 114-Structure Programming Lab

Problem: 1

Write a program in C to find the length of a string without using library functions.

```
#include <stdio.h>

int main()
{
    char str[6] = {'a', 'b', 'c', 'd', 'e', '\0'};
    int i = 0;

    while (str[i] != '\0')
    {
        i++;
    }
    printf("Length of the string is: %d", i);
    return 0;
}
```

Problem: 2

Write a program in C to copy one string into another string and print the second string with the length of characters.

```
#include <stdio.h>

int main() {
    char str[100], cpy[100];
    int i = 0;
    printf("Enter the string.\n");
    fgets(str, sizeof(str), stdin);
    while(str[i] != '\0')
    {
        cpy[i] = str[i];
        i++;
    }
    cpy[i] = '\0';
    printf("The copied string is %s and length of the string is : %d\n",
    cpy, i-1);
}
```

```
return 0;}
```

Problem: 3

Write a program in C to count the number of vowels and consonants in a string.

```
#include <stdio.h>

int main() {
    char str[100], cpy[100];
    int i = 0, vow = 0, cons = 0;
    printf("Enter the string.\n");
    fgets(str, sizeof(str), stdin);
    while(str[i] != '\0')
    {
        if(str[i] == 'a' || str[i] == 'e' || str[i] == 'i' || str[i] ==
'o' || str[i] == 'u' || str[i] == 'A' || str[i] == 'E' || str[i] == 'I'
|| str[i] == 'O' || str[i] == 'U')
            vow++;
        else
            cons++;
        i++;
    }
    printf("Total number of vowels in the string: %d\n", vow);
    printf("Total number of vowels in the string: %d\n", cons);
    return 0;
}
```

Problem: 4

Write a program in C to print the following output.

```
C
CP
CPr
CPro
CProg
CProgr
CProgra
CProgram
CProgramm
CProgrammi
CProgrammin
CProgramming
CProgramming
CProgrammin
CProgrammi
CProgramm
CProgram
CProgra
CProgr
CProg
CPro
CPr
CP
C
```

```
#include <stdio.h>

int main() {
    char str[100] = "CProgramming";
    int c , d;
    for(c=0; c<=11; c++)
    {
        d = c+1;
        printf("%.*s\n", d, str);
    }
    for( c = 11; c>=0; c--)
    {
        d = c+1;
        printf("%.*s\n", d, str);
    }
    return 0;
}
```

Problem: 5

Write a program in C to take three names of an employee and add three names into one string called name.

```
#include <stdio.h>

int main() {
    char name[100], F_name[50], S_name[50], T_name[50];
    int i, j, k;
    printf("Enter first name: ");
    scanf("%s", F_name);
    printf("Enter second name: ");
    scanf("%s", S_name);
    printf("Enter third name: ");
    scanf("%s", T_name);
    for(i = 0; F_name[i] != '\0'; i++)
        name[i] = F_name[i];
    name[i] = ' ';
    for(j = 0; S_name[j] != '\0'; j++)
        name[i+j+1] = S_name[j];
    name[i+j+1] = ' ';
    for(k = 0; T_name[k] != '\0'; k++)
        name[i+j+k+2] = T_name[k];
    name[i+j+k+2] = '\0';
    printf("%s", name);
    return 0;
}
```

Problem: 6

Write a program in C to sort a list of strings in alphabetical order.

```
#include <stdio.h>
#include <string.h>

int main() {
    int n;
    printf("Enter the number of strings.\n");
    scanf("%d", &n);
    char str[n][20], dummy[20];
}
```

```

printf("Enter the list of strings.\n");
for(int i = 0; i<n; i++)
    scanf("%s", str[i]);
for(int i = 0; i<n; i++)
{
    for(int j = i+1; j<n; j++)
    {
        if(strcmp(str[i], str[j]) > 0)
        {
            strcpy(dummy, str[j]);
            strcpy(str[j], str[i]);
            strcpy(str[i], dummy);
        }
    }
}
printf("Sorted in Alphabetical order.\n");
for(int i = 0; i<n; i++)
    printf("%s\n", str[i]);

return 0;
}

```

Problem: 7

Write a program in C to print individual characters of a string in reverse order.

```

#include <stdio.h>
#include <string.h>
#include<stdlib.h>

int main() {
    char str[100];
    int l, i;
    printf("Print individual characters of string in reverse order:\n");
    printf("Input the string : ");
    fgets(str, sizeof(str), stdin);
    l = strlen(str); // Calculate the length of the string
    printf("The characters of the string in reverse are : \n");
    for (i = l - 1; i >= 0; i--) {

```

```
        printf("%c ", str[i]);  
    }  
    return 0;  
}
```

Problem: 8

Write a program in C to count the total number of words in a string.

```
#include <stdio.h>  
#include <string.h>  
  
int main() {  
    char str[100];  
    int len, i, wrd = 0;  
  
    printf("Input the string : ");  
  
    fgets(str, sizeof(str), stdin);  
  
    //len = strlen(str);  
  
    for (i = 0; str[i] != '\0'; i++) {  
        if(str[i] == ' ' || str[i] == '\n' || str[i] == '\t')  
            wrd++;  
    }  
    printf("The number of words in the string: %d", wrd);  
  
    return 0;  
}
```

Problem: 9

Write a program C Program to Check for Palindrome String.

```
#include <stdio.h>
#include <string.h>

int main() {
    char str[100], rev[100];
    int len, flag = 0, j = 0;

    printf("Input the string : ");

    scanf("%[^\\n]s", str);

    len = strlen(str);
    for (int i = len-1; i >= 0; i--)
    {
        rev[j] = str[i];
        //printf("%c", rev[j]);
        j++;
    }
    rev[j] = '\\0';
    for(int i = 0; i<len; i++)
    {
        if(str[i] != rev[i])
        {
            flag = 1;
            break;
        }
    }
    if(flag == 0)
        printf("The string is palindrome.");
    else printf("The string is not palindrome.");
    return 0;
}
```

Problem: 10

Write a program in C to compare two strings without using string library functions.

```
#include <stdio.h>
#include <string.h>

int main() {
    char str1[100], str2[100];
    int len, flag = 0, j;

    printf("Input the first string : ");
    fgets(str1, sizeof(str1), stdin);

    printf("Input the second string : ");
    fgets(str2, sizeof(str2), stdin);

    len = strlen(str1);
    for (int i = 0; i < len; i++)
    {
        if(str1[i] != str2[i])
        {
            flag = 1;
            break;
        }
    }
    if(flag == 0)
        printf("The strings are equal.");
    else printf("The strings are not equal.");
    // Using library function
    // j = strcmp(str1, str2);
    // if(j > 0) printf("The unmatched character of string1 is greater
than string2.");
    // else if(j < 0) printf("The unmatched character of string2 is
greater than string1.");
    // else printf("The strings are equal.");
    return 0;
}
```

Problem: 11

Write a program in C to count the total number of alphabets, digits and special characters in a string.

```
#include <stdio.h>
#include <string.h>

int main() {
    char str[100];
    int len, alph = 0, dig = 0, spch = 0;

    printf("Input the string : ");
    fgets(str, sizeof(str), stdin);

    len = strlen(str);
    for(int i = 0; i<len; i++)
    {
        if((str[i] >= 'a' && str[i] <= 'z') || (str[i] >= 'A' && str[i]
<= 'Z'))
            alph++;
        else if (str[i] >= '0' && str[i] <= '9')
            dig++;
        else
            spch++;
    }
    printf("Number of Alphabets in the string is : %d\n", alph);
    printf("Number of Digits in the string is : %d\n", dig);
    printf("Number of Special characters in the string is : %d\n", spch);
    return 0;
}
```